

Sai Praneet Reddy Chinthala

+1 571-363-5378 | praneetreddy66@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

George Mason University

M.S. Data Analytics Engineering – GPA: 3.97/4.0

Fairfax, Virginia
Aug 2024 – May 2026

Mahindra University

B.Tech. Electronics and Computer Engineering

Hyderabad, India
Sept 2020 – May 2024

SKILLS & TECHNOLOGIES

ML & GenAI: RAG Pipelines, LLMs, Prompt Engineering, NLP, Sentence Transformers (SBERT), FLAN-T5, OpenAI API, FAISS, BM25, LangChain, Cohere Rerank, **Langfuse** (LLM Observability), PyTorch, TensorFlow, Scikit-learn, **XGBoost**

Data Engineering & Analysis: Python, SQL, Feature Engineering, Data Cleansing, Exploratory Analysis, Pandas, NumPy, Apache Spark, **PySpark**, **ETL/ELT Pipelines**, MySQL, MongoDB

Cloud & DevOps: AWS (S3, EC2, Lambda), Azure, **Docker**, CI/CD (GitHub Actions), Git, GitHub

Visualization & Reporting: Matplotlib, Seaborn, Jupyter Notebook, Databricks SQL

Languages: Python, SQL, R, Java, JavaScript

Coursework: Probability & Statistics, DBMS, Big Data, Machine Learning, Data Science, Distributed Systems

WORK EXPERIENCE

Graduate Teaching Assistant (AIT 614 – Big Data Essentials)

George Mason University

Jan 2026 – May 2026

Fairfax, Virginia

- Led 10 hands-on labs across **two sections** of a graduate Big Data course, spanning **Databricks**, **Apache Spark/PySpark**, **MongoDB**, **Hadoop**, and **AWS** cloud, plus an introduction to **LangChain-based RAG**.
- Graded **100+ Python and PySpark submissions** across **40+ students** on ETL, model training, and **feature engineering**, with targeted debugging feedback.
- Verified and corrected course labs for technical accuracy (solutions used as the student reference), and mentored student teams through their term-long big-data projects on scoping, tooling, and debugging.

Generative AI Engineer Intern

Mahindra & Mahindra Ltd

Feb 2024 – May 2024

Remote

- Consolidated **20+ raw machine-breakdown (“why-why”) root-cause Excel logs** from 3 plants into one standardized dataset using **Python (pandas/openpyxl)**, resolving inconsistent formatting, encoding errors, and merged/blank cells.
- Adapted an existing internal **RAG chatbot** (OpenAI embeddings + vector search) to the cleaned corpus – generating document embeddings and validating retrieval so engineers could query historical breakdown fixes in natural language.
- Tuned **data preparation** and prompt formatting to improve answer relevance for plant-maintenance queries.

UI Technologies Intern

Aspire InfoLabs

Jun 2023 – Jul 2023

Hyderabad, India

- Won a **university-wide hackathon** hosted by **Aspire InfoLabs** (100+ participants); **MentorMe**, a student-mentor matching concept, won with a 5-person team.
- Built the backend authentication service on **Node.js/Express/MongoDB** – **Google OAuth** login, **JWT** sessions, and email **OTP** password reset (Nodemailer).

PROJECTS

DAPSE: Arctic Policy Intelligence Engine | Master’s Capstone (team of 6) | Product Owner

Jan 2026 – May 2026

- As Product Owner, co-developed a **RAG engine** with **BM25 + FAISS hybrid retrieval**, Reciprocal Rank Fusion, and **Cohere Rerank** over a **955+ document corpus**, reaching **nDCG@5 = 0.832** and **Precision@5 = 0.954**.
- Directed delivery of a **verify-then-escalate** agent loop and an **8-dimension geopolitical risk-scoring engine**, with **Langfuse** observability, an automated evaluation suite, and **Docker** deployment.

Bridge Load-Posting Prediction | GitHub

Aug 2025 – Dec 2025

- Built a **PySpark ETL + ML pipeline** on **950K+ National Bridge Inventory records**; engineered **15+ features** (interaction, polynomial, binned) and handled a 9:1 class imbalance with class weighting.
- Benchmarked logistic-regression, decision-tree, and **XGBoost** models; tuned XGBoost to **0.94 ROC-AUC** (vs. 0.91 baseline) and persisted predictions to **MongoDB** with automated **Databricks SQL** reporting.

Healthcare Financial Analysis | GitHub

Aug 2024 – Dec 2024

- Built a multi-tool **ETL pipeline** (Python, MySQL, R) on **4,233 CMS Hospital Cost Report records**; applied **regression and correlation analysis** to quantify the charity-care vs. net-income tradeoff across U.S. hospitals.